CSE-3523: Microprocessors, Microcontrollers & Embedded Systems

Midterm Syllabus Summary (Spring-26)

Segment 1

Focus Area:

- Introduction to Microprocessor and its main tasks
- Microcontroller vs microprocessor
- evolutions of Microprocessor and its types
- Hardware Components of a Microprocessor
- Function of buses,System Bus,Expansion type Buses
- Instruction Execution Cycle

Segment 2

This segment includes the **entire content** of Segment 2 from the Lecture material.

Segment 3 (Part-1)

Focus Area:

- Assembly Language
- Assembly Language syntax :Rules,Programe Statement,Program data(byte variable ,word variable) program structure,memory model,procedure

Segment 3 (Part-2)

Focus Area:

- Data Transfer Instructions: MOV, PUSH, POP
- Arithmetic Instructions: ADD, INC, ADC, MUL
- Logical Instructions: AND, OR, XOR,SHIFT
- Program control Instruction:JUMP,LOOP (Example:Writing assembly language program using jump instruction)

focus on understanding the concepts rather than rote memorization. Practice writing programs and solving problems involving ***arithmetic, logical, Series calculation using JMP instruction and data transfer instructions.as well as addressing modes, registers, and flags****. Clear understanding and regular practice will help you answer questions accurately and score higher marks*.